

Annamalai Kalyanasundaram

Full-Stack GenAI Engineer | RAG · Agentic AI · Python · FastAPI | AWS Solutions Architect

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SUMMARY

Software engineer with ~2 years of production experience and a focused specialization in generative and agentic AI. Builds LLM applications with LangGraph, LangChain, and CrewAI — including RAG pipelines, FastAPI microservices, Docker/Kubernetes deployment, and evaluation via RAGAS and LangSmith. Delivers AWS-backed services, REST APIs, and CI/CD with measurable operational impact. AWS Certified Solutions Architect – Associate.

TECHNICAL SKILLS

GenAI / LLM: LangGraph, LangChain, CrewAI, RAG (hybrid, corrective, Self-RAG), FAISS, ChromaDB, Pinecone, Milvus, RAGAS, LangSmith, Hugging Face, Transformers, PEFT (LoRA/QLoRA), prompt engineering, guardrails

Languages: Python, JavaScript / Node.js, SQL, Bash, Kotlin

Backend & APIs: FastAPI, Express.js, REST microservices, request validation, structured error handling

Cloud & DevOps: AWS (EC2, VPC, IAM, Lambda, ECS, S3, RDS, CloudWatch), Docker, Kubernetes, GitHub Actions, Jenkins, Git

Data & Frontend: MySQL, document ingestion for RAG, Neo4j; React.js; Jetpack Compose

EXPERIENCE

Software Developer

Jun 2024 – Jul 2026

Arche Global Pvt Ltd — Bengaluru, India

- Improved operational efficiency by ~30% on Pulse, a hybrid-cloud operations platform, by automating monitoring workflows and streamlining release processes.
- Designed and maintained 15+ production REST APIs (Node.js, Express.js) supporting asset management, ticketing, and monitoring under high-frequency traffic.
- Built CI/CD pipelines (GitHub Actions, Jenkins) that reduced deployment friction across staging and production; containerized services with Docker on AWS (EC2, VPC, IAM, ECS, Lambda).
- Optimized MySQL schemas and queries for high-frequency operational data; delivered React.js dashboards for real-time infrastructure visibility.
- Partnered with DevOps and infrastructure teams in Agile sprints; contributed to code reviews and reliable releases.

SELECTED PROJECTS

Agentic RAG Advisor

Capstone · Oct 2026

LangGraph · LangChain · FastAPI · Docker · Kubernetes · ChromaDB / FAISS · RAGAS · LangSmith

- Built an end-to-end agentic RAG service for a regulated-domain advisor covering ingestion, chunking, embeddings, vector retrieval, prompt assembly, guardrails, and grounded generation.
- Orchestrated multi-step reasoning with LangGraph (state, memory, tool use, human-in-the-loop) and served capabilities through FastAPI microservices with validation and structured errors.
- Instrumented workflows with LangSmith tracing and a RAGAS evaluation harness; packaged with Docker, deployed to Kubernetes, and automated CI with GitHub Actions.

RAG Assistants & Agentic Workflows

Capstone · Oct 2026

LangChain · CrewAI · Vector databases · Open-source & API LLMs · FastAPI

- Delivered a domain-aware chatbot with model switching (Mistral and API LLMs) and a RAG support assistant over internal knowledge bases.
- Implemented an agentic market-insight workflow with tool calling, routing, memory, and structured outputs.

ADDITIONAL PROJECTS

BIAL Plus — Internal Employee Mobile App

Production · Google Play

Kotlin · Jetpack Compose · Azure AD · Microsoft Intune · CI/CD

- Shipped Bangalore International Airport’s official internal mobility platform for HCM, shift management, and HR services used in day-to-day airport operations.
- Integrated Azure AD single sign-on and Microsoft Intune (MDM/MAM); diagnosed and resolved critical authentication blockers to maintain 24/7 availability.

Pulse — Cloud Operations Platform

Arche Global

React.js · Node.js · Express.js · MySQL · AWS · Docker

- Architected the central hybrid-environment operations tool for infrastructure visibility and incident response.
- Implemented 15+ REST APIs and a MySQL-backed data layer with optimized queries; built real-time React monitoring dashboards for cloud assets and tickets.

ArcheOne — Productivity Platform (Android)

Arche Global

Kotlin · Android SDK · REST APIs · Push Notifications

- Delivered end-to-end product features including Business Card, SOS, Deskart, TravelDesk, and Helpdesk.
- Integrated real-time push notifications and optimized REST clients for reliable performance across a wide range of Android devices.

EDUCATION

Bachelor of Technology (B.Tech), Engineering

2021

Sri Venkateswara College of Engineering, Chennai

Advanced Generative and Agentic AI

Completed Oct 2026

Naresh Technologies

- **Foundations:** Python for AI, neural networks, PyTorch, Transformer architecture, Hugging Face, multimodal LLMs, prompt engineering and guardrails.
- **RAG & agents:** Chunking, embeddings, Pinecone/FAISS/ChromaDB/Milvus, GraphRAG, LangGraph, LangChain, CrewAI, MCP, LangSmith, RAGAS evaluation.
- **Serving:** PEFT fine-tuning (LoRA/QLoRA), quantization, vLLM.

CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate** — valid through March 2027
- **APTECH Certified Python Developer**

LANGUAGES

English (Fluent) · Hindi (Fluent) · Tamil (Native)